

FitBuddy — Project Documentation

Project: FitBuddy – AI Fitness Plan Generator Using Gemini

Purpose: Project-track submission material adapted to the supplied AI/ML/Gen-AI template.

Abstract

FitBuddy is an AI-assisted fitness planning web application that uses profile information and Google Gemini to generate a personalized weekly workout plan and related nutrition tips.

Objectives

Simplify initial workout planning, support goal/intensity-based personalization, save plan information, and allow feedback-based revision.

Methodology

The browser collects user details; the backend validates and routes requests; the Gemini integration generates content; and the database layer stores relevant records.

Technology

Python, FastAPI, Jinja2, HTML/CSS/JavaScript, SQLAlchemy, SQLite or configured SQL database, Pydantic, and Google Gemini SDK, as described in the supplied project documentation.

Limitations

Outputs depend on model availability and quality. AI-generated exercise and nutrition suggestions may be inaccurate or unsuitable for some users. Users should consult qualified professionals where appropriate.

Future scope

Add automated tests, structured output validation, stronger authentication/authorization, improved observability, accessibility testing, deployment automation, and user feedback evaluation.